DESIGN OF 6 LANE FLEXIBLE PAVEMENTS BY USING IIT PAVE SOFTWARE

A Major Project Report Submitted in Partial Fulfilment of the requirement for the award of the Degree of

BACHELOR OF TECHNOLOGY

IN

CIVIL ENGINEERING

Submitted

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MALLA REDDY INSTITUTE OF TECHNOLOGY

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CERTIFICATE

This is to certify that project entitled **"DESIGN OF 6 LANE FLEXIBLE PAVEMENT BY USING IIT PAVE SOFTWARE"** submitted by T. HARSHAVARDHAN (19RJ5A0161) in partial fulfilment for the award of the 'Bachelor of Technology' in the Civil Engineering Department to Malla Reddy Institute of Technology, Maisammaguda, Secunderabad, Telangana State.

The results emboided in this project have not been submitted to any other University or Institution for the award of any Degree or Diploma.

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We hereby declare that the contents presented in the project entitled **"DESIGN OF 6 LANE FLEXIBLE PAVEMENTS BY USING IIT PAVE SOFTWARE"** submitted in partial fulfilment for the award of Bachelor of Technology degree to **MALLA REDDY INSTITUTE OF TECHNOLOGY** is a record of original work investigated by me under the supervision of **T. NARESH.** This is to state that the results emboided in this project have not been submitted to any other University or Institution for the award of Degree or Diploma.

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ABSTRACT

Road transport plays a vital role in social, economic and industrial development of the country. Roads are used by different vehicle class which results in failure of the pavement due to early deterioration. The failure of bitumen pavement is due to fatigue cracking and rutting deformation. This paved a way for developing the software like IIT PAVE which determines the stress and strain values at critical locations at different layer of pavement. This project aims at collection of data related to traffic studies and subgrade soil CBR values from the field and designing the flexible pavement as per the guidelines of IRC: 37-2018. Later it is analysed using IIT PAVE Software to check whether it satisfies the specific requirements. This project briefly describes the use of IIT PAVE Software for project analysis of National Highway NH365BB (Suryapet - Khammam) with a chainage length of 10km from Mothey, Suryapet to Nayakulagudem, Kothagudem, Telangana.

Keywords: Fatigue, rutting, IIT PAVE, IRC, NH365BB.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

The design of flexible pavement depends on different factors that affect the strength and CBR value of the subgrade soil such as load, properties of soil, environmental and climatic factors. Using flexible pavement is necessary to ensure road safety in a large city and will help to save time and avoid difficulties and errors which might happen during the design. Pavements are constructed as per IRC guidelines. The main reason for failure of bitumen pavement is due to excessive strain and deformation at critical locations in the pavement.

1.2 DEFINITION OF ROAD

A wide way leading from one place to another, especially one with a specially prepared surface which vehicles can use.

1.3 COMPOSITION OF ROAD STRUCTURE

Road Structure Cross Section is composed of the following components

1. Sub Base
2. Base Course
3. Sub Grade
4. Surface/Wearing Course

1.3.1 Sub Bases

It is layer of granular material provided above subgrade generally natural gravel. It is usually not provided on subgrade of good quality.

a. Function of Sub base in Road Cross Section

- It enables traffic stresses to be reduced to acceptable levels in subgrade in the Road Cross Section.
- It acts as a working plate form for the construction of upper pavement layers.
- Acts as a drainage layer, by protecting the subgrade from wetting up.
- It intercept upward movement of water by capillary action.

- It acts as a separating layer b/w subgrade and road base. By this it prevent the two layers from mixing up.

b. Characteristics of materials used in Sub Base:

The subgrade material should be clean and free from organic matter and should be able to be compacted by roller, to form stable sub-base. The material should have the following characteristics:

- Well graded, uniformity coefficient (D_{60}/D_{10}) should not be less than 3.
- Fraction passing sieve 200 shall not be greater than 2/3rd of the fraction passing sieve 40.
- Should have a L.L not greater than 25% & P.I not greater than 6 .
- CBR should not be less than 25.
- In coarse grain, aggregate retained by 10 sieve, %age of wear shall not be greater than 5%.
- The maximum dia of any particle shall not be greater than 2/3 rd of the layer thickness of sub-base.

Typical particle size distribution for the sub-base (granular) which will the meet strength requirements are:

Table 1.1: Sieve Analysis Values

I.S Sieve Size	% By mass of total Aggregates passing test sieve
50	100
37.5	80-100
20	60-100
5	30-100
1.15	170-75
0.3	9-50
0.075	5-25

D_{15} (sub base) < 5

* To avoid intrusion of silt and clay material in sub-base from subgrade.

1.3.2 Sub Grade in Road Structure Cross Section

D_{15} (sub grade)

Recommended plasticity characteristic for granular Sub Base are;

Table 1.2: Liquid Limit & Plasticity Index Values

Climate	Liquid Limit (L.L)	Plasticity Index (P.I)
Moist or wet tropical	< 35	< 6
Seasonal wet tropical	< 45	< 12
Arid & Semi Arid	< 55	< 20

1.3.3 Base Courses in Road Structure Cross Section

- It is the layer immediately under the wearing surface (Applied whether the wearing surface is bituminous or cement concrete and whether its a thick or thin bituminous layer).
- As base course lies close under the pavement surface it is subjected to severe loading. The material in a base course must be of extremely high quality and its construction must be done carefully. The LA Abrasion test can determine the quality of the aggregate for this purpose.

1.3.4 Types of Base Course

1. **Granular Base Course:**

It is a mixture of soil particles ranging in size from coarse to fine. Processing involves crushing oversized particles and screening where it is necessary to secure the desired grading. The requirements of a satisfactory soil aggregate surface are;

- Stability,

- Resistance to abrasion,
- Resistance to penetration of water,
- Capillary properties to replace moisture lost by surface evaporation upon the addition of wearing course requirement change.

2. **Macadam Base:**

Successive layers of crushed rock mechanically locked by rolling and bonded by stone screening (rock duct, stone chips etc).

3. **In-water bound Macadam:**

The crushed stones are laid, shaped and compacted and then finer materials are added and washed into surface to provide a dense material.

4. **Treated Bases:**

Compose of mineral aggregate and additive to make them strong or more resistant to moisture. Among the treating agents is bitumen.

1.3.5 Surface/Wearing Course in pavement cross section

The top layers of pavement which is in direct contact with the wheel of the vehicle. Usually constructed of material in which bitumen is used as binder materials.

a. Bituminous Pavement:

Consists of combination of mineral aggregate with bituminous binder ranging from inexpensive surface treatment ¼ in or less thick to asphaltic concrete. For good service throughout the full life, bituminous pavement must retain the following qualities:

- Freedom from cracking or raveling.
- Resistance to weather including the effect of surface water heat and cold.
- Resistance to internal moisture, particularly to water vapors.
- Tight impermeable surface or porous surface (if either is needed for contained stability of underlying base or subgrade).
- Smooth riding and non skidding surface.

The design should be done so that to meet the above requirements for considerable number of years (need proper design and construction supervision). Pavements meeting all the requirements above can be produced if these construction processes are followed:

- Heat a viscous bituminous binder to make it fluid. Then in a plant, mix it with heated aggregate. Place and compact the mixture while it is hot.
- Use fluid bituminous binder. Mix it with aggregate at normal temperature. Mixing may be done at a plant (plant mix) or on the prepared roadway base (road mix). Spread and compact the mixture at normal temperature.
- Add solvent such as naphtha or kerosene to a viscose bituminous binder to make it fluid with aggregate at normal temperature by either plant or road mix methods. Spread and compact at normal temperature before solvent evaporates.
- Use fluid emulsion of viscous bituminous binder in watee. Mix it with aggregate at normal temperature by either plant or road mix method. Spread and compact at normal temperature before the emulsion breaks down with its components.
- Spread and compact clean crushed aggregate as for water bound macadam. Over it, spray heated dissolved or emulsified bituminous binder which penetrates open areas of the rock and binds the aggregate together. Thus is commonly called “Penetration Method”.
- Spread bituminous binder over the roadway surface then cover it with properly selected aggregate. This is commonly called the “Inverted Penetration Method”.

Selections based on the requirements and economy, large volume of heavy vehicles, low traffic volume etc.

1.4 TYPES OF PAVEMENTS

1.4.1 Pavement Definition

Pavement is the actual travel surface especially made durable and serviceable to withstand the traffic load commuting upon it. Pavement grants friction for the vehicles thus providing comfort to the driver and transfers the traffic load from the upper surface to the natural soil.

In earlier times before the vehicular traffic became most regular, cobblestone paths were much familiar for animal carts and on foot traffic load. Pavements are primarily to be used by vehicles and pedestrians. Storm water drainage and environmental conditions are a major concern in the designing of a pavement. The first of the constructed roads date back to 4000 BC and consisted of stone paved streets or timber roads. The roads of the earlier times depended solely on stone, gravel and sand for construction and water was used as a binding agent to level and give a finished look to the surface. All hard road pavements usually fall into two broad categories namely

a. Flexible Pavement,

b. Rigid Pavement.

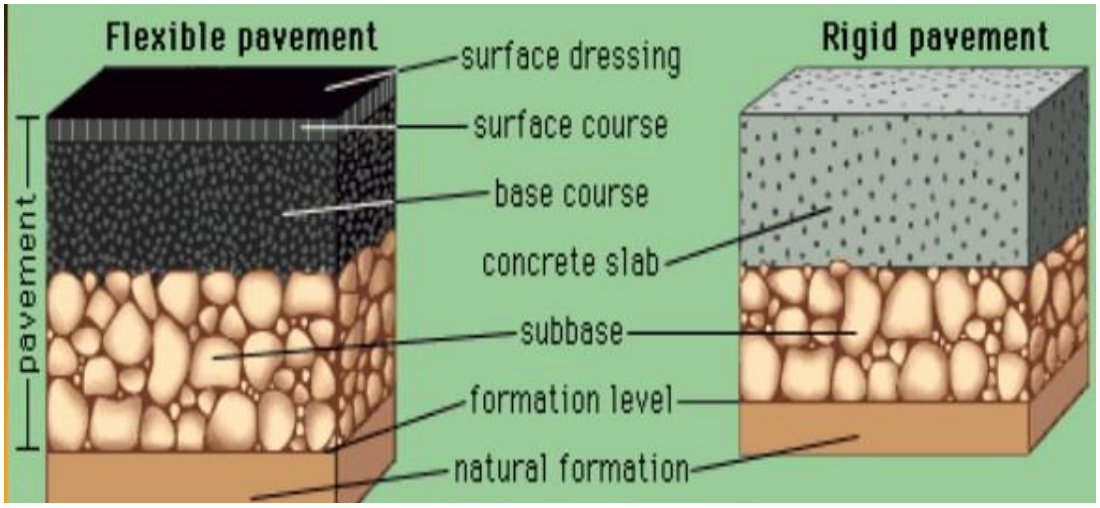


Figure 1.1 typical-detail-of-road-section

1.4.2 Flexible Pavement

Flexible pavements are which reflect the deformation of subgrade and the subsequent layers to the surface. Flexible, usually asphalt, is laid with no reinforcement or with a specialized fabric reinforcement that permits limited flow or repositioning of the roadbed under ground changes.

- The design of flexible pavement is based on load distributing characteristic of the component layers. The black top pavement including water & gravel bound macadam fall in this category.
- Flexible pavement on the whole has low or negligible flexible strength flexible in their structural action). The flexible pavement layers transmit the vertical or compressive stresses to the lower layers by grain transfer through contact points of granular structure.

- The vertical compressive stress is maximum on the pavement surface directly under the wheel load and is equal to contact pressure under the wheels. Due to the ability to distribute the stress to large area in the shape of truncated cone the stresses get decreased in the lower layer.

- As such the flexible pavement may be constructed in a number of layers and the top layer has to be strongest as the highest compressive stresses.

- To be sustained by this layer, in addition to wear and tear, the lower layer have to take up only lesser magnitude of stress as there is no direct wearing action die to traffic loads, therefore inferior material with lower cast can be used in the lower layers.

1.4.3 Rigid pavement

The rigid characteristic of the pavement are associated with rigidity or flexural strength or slab action so the load is distributed over a wide area of subgrade soil. Rigid pavement is laid in slabs with steel reinforcement.

- The rigid pavements are made of cement concrete either plan, reinforced or prestressed concrete.

- Critical condition of stress in the rigid pavement is the maximum flexural stress occurring in the slab due to wheel load and the temperature changes.

- Rigid pavement is designed and analyzed by using the elastic theory.

1.4.4 Difference Between Flexible And Rigid Pavement

- Flexible pavement differ from rigid pavement in terms of load distribution. In flexible pavements load distribution is primarily based on layered system. While, in case of rigid pavements most of the load carries by slab itself and slight load goes to the underlying strata.

- Structural capacity of flexible pavement depends on the characteristics of every single layer. While, the structural capacity of rigid pavements is only dependent on the characteristics of concrete slab. This is so, because of low bearing soil capacity of underlying soil.

- In flexible pavements, load intensity decreases with the increase in depth. Because of the spreading of loading in each single layer. While, in case of rigid pavement maximum intensity of load carries by concrete slab itself, because of the weak underlying layer.
- In flexible pavement deflection basin is very deep, because of its dependency on the underlying layers. While in case of rigid pavement, deflection basin is shallow, this is because of independency of rigid pavement on the underlying layers.
- Flexible pavement has very low modulus of elasticity (less strength). Modulus of elasticity of rigid pavement is very high, because of high strength concrete and more load bearing capacity of the pavement itself. Than compared to flexible pavements.
- In flexible pavements, underlying layers play very important role. Therefore, more role are playing only underlying layers. In case of rigid pavements, slight function of underlying layers. Maximum role is playing by the top layer (that is slab) by itself. Therefore, minute part is taking by sub layers.

1.4.5 Requirements of a pavement:

- Sufficient thickness to distribute the wheel load stresses to a safe value on the sub grade soil.
- Structurally to strong to withstand all types of stress imposed upon it.
- Adequate coefficient of friction to prevent skidding of vehicles.
- Smooth surface to provide comfort to road users even at high speed.
- Produce least noise from moving vehicles.
- Dust proof surface so that traffic safety is not impaired by reducing visibility.
- Impervious surface, so that sub-grade soil is well protected and long design life with low maintenance cost.

1.5 CONSTRUCTION OF MACADAM ROADS

1.5.1 WBM (Water Bound macadam) roads construction

The water bound macadam road construction technique was given by the John Macadam. This technique in present day we are used three materials:

a. Aggregates b. Screeners c. *Binders.*

a. Aggregates:

Aggregates are using the different grades in WBM. IRC(Indian Roads Congress) has classified the coarse aggregates into 3 grades, according to their size.For the construction of the WBM roads aggregates are used in the sub-base, base and surface course and so the aggregates are divided into 3 grades according to their size. Grade 1 - particles of size 90 mm to 40 mm. Grade 2 - particles of size 63 to 40 mm. Grade 3 - particles of size 50 to 20 mm.

The grade 1 aggregates having size of 90 mm to 40 mm are preferred for the sub-base material and grade 2 for the base and grade 1 for the surface course. However, if we only use the WBM as the surface course, it gets deteriorated fast due to abrasion with the traffic so, bituminous surfacing over the WBM is general practice.

b. Screeners:

Screeners, are the aggregates of the smaller sizes, generally 12.5 mm or 10 mm, for grade A and grade B. They are of the same chemical composition as of the coarse aggregates. For economic considerations IRC has suggested non plastic materials such as, crushed over burnt bricks, moorum, gravels, etc. provided the liquid limit of the material is less than 20%, plasticity index is less than 6.0% and the portion of fines passing 0.075 mm sieve is less than 10%. However if crush-able type of aggregates are used, use of the screeners may be disposed off.

c. Binders:

Binders, are the layers of materials which are laid after the compaction of the aggregates and the screening materials one after the another. Kankar dust or lime stone dust may be utilized if locally available. The binding material with plasticity index value of 4% to 9% is used in surface course construction; the plasticity index of binding course material should be less than 6% in the case of the WBM layers used as base course or sub-base course, with bituminous surfacing. However if the screening used are of crushable material like moorum or soft gravel, there is no need to apply binding material, unless the plasticity index value is low.

1.5.2 WMM(Wet mix macadam) road construction

Aggregates used are of the smaller sizes, varies between the 4.75 mm to 20 mm sizes and the binders(stone dust or quarry dust having PI(Plasticity Index) not less than 6%) are premixed in a batching plant or in a mixing machine. Then they are brought to the site for overlaying and compaction.

The PI(plasticity Index) of the binding material is kept low because it should be a sound and non plastic material. If the plasticity index is more then there are the chances of the swelling and more water retention properties. So this value should be kept in mind.

1.5.3 Comparison of the WBM and WMM road construction

Although the cost of construction of the WMM is said to be more than that of the WBM sub-base and bases but the advantages given below will compensate for that. Here are the points of difference:

1. The WMM roads are said to be more durable.
2. The WMM roads gets dry sooner and can be opened for traffic withing less time as compare to the WBM roads which take about one month for getting dry.
3. WMM roads are soon ready to be black topped with the Bituminous layers.
4. WMM roads are constructed at the faster rate.
5. The consumption of the water is less in case of the WMM roads.
6. Stone aggregates used in WBM is larger in size which varies from 90 mm to 20 mm depending upon the grade but in case of the WMM size varies from 4.75 mm to 20 mm.
7. In case of WBM, stone aggregates, screenings and binders are laid one after another in layers while in WMM, aggregates and binders are pre-mixed in the batching plants and then brought to the site for overlaying and compacting.
8. Materials used in the WBM are the stone aggregates, screenings and binder material(Stone dust with water) while in WMM material used are only stone aggregates and binders.
9. Quantity of the WBM is generally measured in cubic meters while that of the WMM in square meters.

1.6 IIT PAVE FOR PAVEMENT ANALYSIS

IIT PAVE software is an improved version of FPAVE which is developed by research scheme R-56 of MORTH. This is a multilayer analysis programmer used for design, analysis of the flexible or bitumen pavement using IRC: 37-2018 guidelines. In this software we enter thickness of pavement layers, loads applied over the surface of pavement, tire pressure, spacing between the wheels and Poisson´s ratio as inputs. After running the software actual horizontal tensile strain and vertical compressive strains at critical locations of the pavement are obtained as output.

Figure 1.2: IIT PAVE

1.7 STUDY AREA

Design of 6 lane pavement for the project analysis of National Highway NH365BB (Suryapeta – Khammam) with chainage length of 10km from Mothey, Suryapet to Nayakulagudem, Kothagudem Telangana.

National Highway 365BB is a national highway in India. It is a secondary route of National Highway 65. NH-365BB traverses the states of Telangana and Andhra Pradesh in India. It starts at Suryapet and ends at Kovvur(Rajamahendravaram). Major cities on this route are Suryapet, Khammam and Rajamahendravaram.

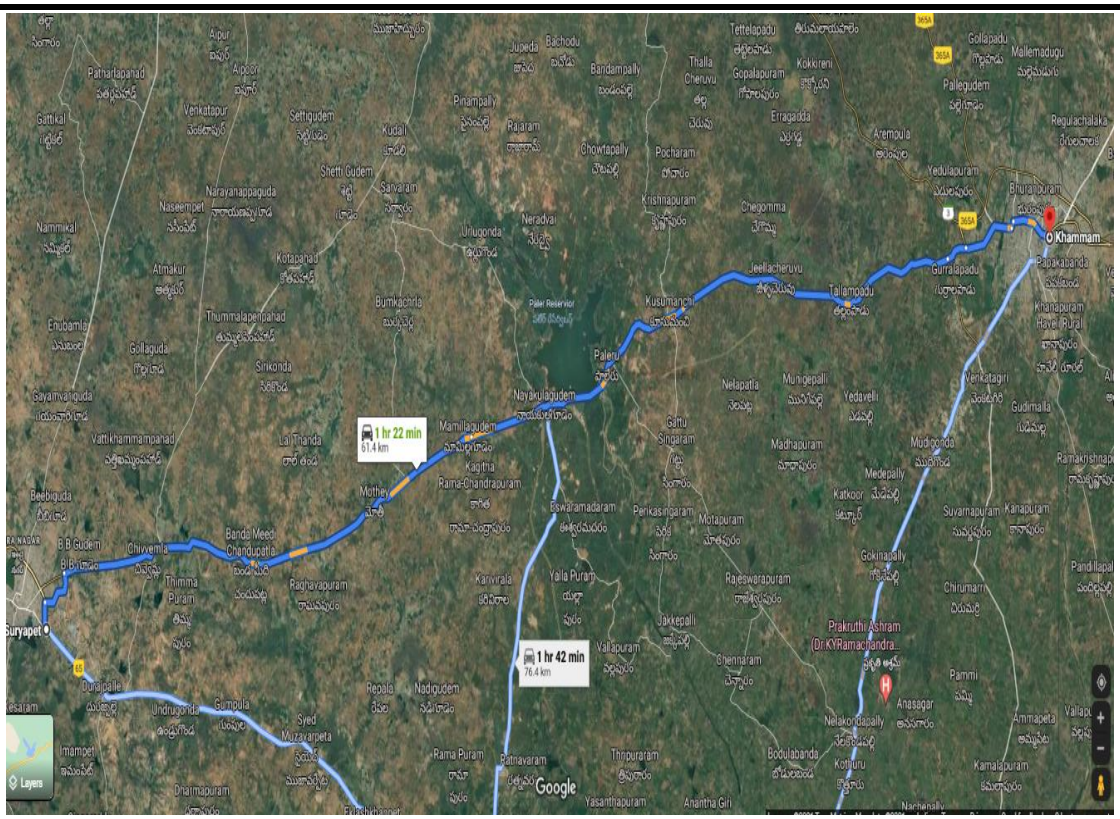


Figure 1.3 Satellite Map for the NH-365BB from Suryapet to Khammam

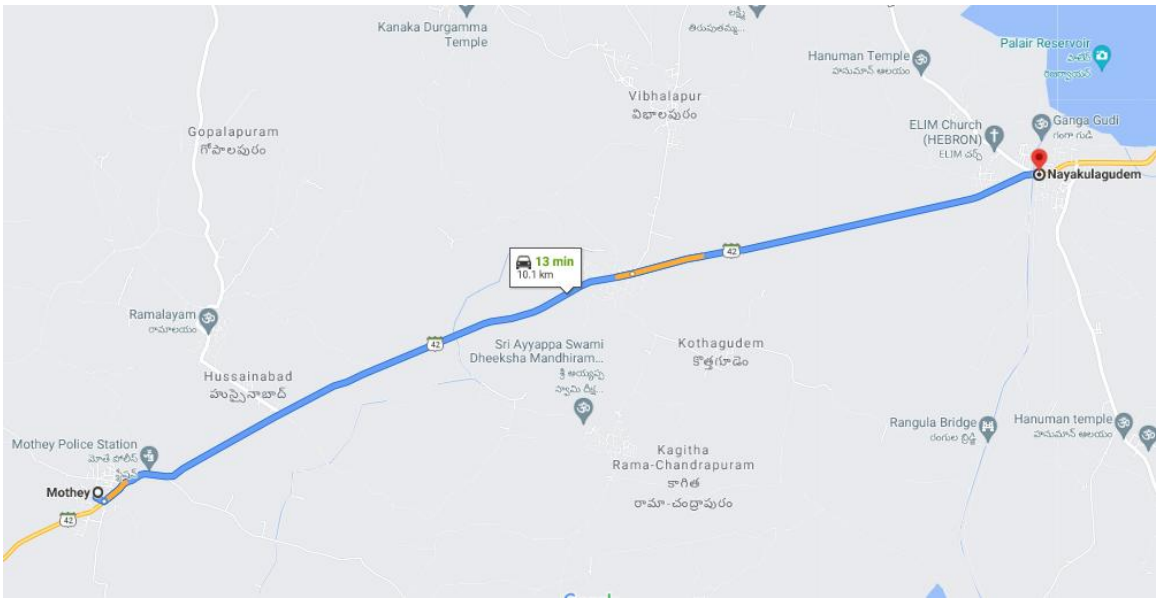


Figure 1.4 Map for Mothey to Nayakulagudem

CHAPTER 2

LITERATURE REVIEWS

Harish G R (4) (2017), "Analysis of Flexible Pavements Design Using IIT PAVE", Imperial Journal of Interdisciplinary Research (IJIR), Vol-3, Issue-6, 2017, ISSN 2454-1362. This paper studies the design of flexible pavement by using cementitious materials for base and sub-base layers for crack relief and cementitious base and sub-base with SAMI (Stress Absorbing Membrane Interface) as per the guidelines of IRC: 37-2012 codebook and later the same pavement is analyzed in the IIT PAVE software. IIT PAVE is the multi-layer analysis programmer used for analysis of flexible pavement and to determine the stresses and strains at critical locations of the pavement. In this case the road stretch is selected around Bangalore and engineering properties of subgrade soil has been studied. From their study it is concluded that the pavement which is designed by using cementitious base and sub-base with SAMI will give better serviceability because of substantial reduction in strains and thickness, and also it is cost effective compared to other pavement materials.

Komershetty Gouthami, Dr KSR Murthy, D. Sandya Rani (7) (2017), "Flexible pavement design and Comparison of alternative Pavements using IRC 37-2012 IIT PAVE", International journal for scientific Research and Development, vol-5, Issue-6, 2017, ISSN: 2321-0613. This paper deals in studying the life cycle cost of different pavement compositions. For this purpose, they have collected the information associated to vehicular traffic studies, axle load survey and sub-grade CBR values of a selected road. The project road considered for study is Solapur to Sangareddy section of NH-9. From the obtained data, the design of flexible pavement is carried out using IRC: 37-2012 guidelines. Later a roadway is analyzed in IIT PAVE software and compared with the pavement which is designed by the alternative materials available in the IRC: 37-2012 with the conventional granular materials. Based on the results obtained in design of pavement using IRC: 37-2012 and analysis carried out using IIT PAVE software, they have concluded that the alternate pavements with less total pavement thickness can cater the same traffic as that of conventional pavement except the alternate pavement with CTB and GSB.

CHAPTER 3

OBJECTIVE OF THE STUDY

1. Traffic Analysis by using collected traffic volume count data by adopting suitable CVPD values.
2. Pavement design based on average daily traffic as per the IRC: 37 -2018 guidelines and analyzing pavement in IITPAVE software.
3. To design and evaluate the performance of fatigue and rutting performance of flexible pavements using IITPAVE Software.
4. To design of various pavement Layer thickness with the help of IITPAVE Software.
5. To determine the best engineering and economic alternatives for the road sections.

CHAPTER 4

METHODOLOGY

The flexible pavement is designed by an empirical method, as per the IRC guidelines (IRC: 37-2018) which is based mainly on CBR value of subgrade soil and design charts to determine the total pavement thickness. The design of flexible pavement is done according to IRC: 37-2018 guidelines and it is analyzed in IIT PAVE software. The road selected for this project is Suryapet – Khammam section of NH-365BB.

A flexible pavement is modelled as an elastic multilayer structure. The mechanistic empirical software called IITPAVE is used to analyze the pavement. The stress analysis IITPAVE software is used to compute the stresses and strains at critical location of the pavement. Horizontal Tensile strain at the bottom of the bitumen layer and Vertical Compressive strain on the top of the subgrade are considered as the critical parameters to limit cracking and rutting in bituminous and non-bituminous layers.

The allowable tensile and compressive strains at critical location of the pavement are calculated using empirical equations mentioned in IRC: 37-2018 guidelines which is later compared with the actual tensile and compressive strains as computed by IIT PAVE software.

CHAPTER 5

EXPERIMENTAL WORK

5.1 FIELD INVESTIGATION OF SOIL

5.1.1 California Bearing Ratio Test

The California bearing ratio test is penetration test meant for the evaluation of subgrade strength of roads and pavements. The results obtained by these tests are used with the empirical curves to determine the thickness of pavement and its component layers. This is the most widely used method for the design of flexible pavement. It is also defined as the ratio of force per unit area required to penetrate a soil mass with standard circular piston at the rate of 1.25 mm/min. to that required for the corresponding penetration of a standard material.

The equipments used are Mould, Steel Cutting collar, Spacer Disc, Surcharge weight, Dial gauges, IS Sieves, Penetration Plunger, Loading Machine, Miscellaneous Apparatus. The procedure of conducting CBR test is as follows:

- First the sample of 5.5 kg is taken. The dry density should be either the field density or if the subgrade is to be compacted, at the maximum dry density value obtained from the Compaction test.
- The optimum moisture content found out from the compaction test is applied in the CBR test. Add the obtained optimum water % to the soil sample.
- Place a spacer disc and filter paper in the mould and soil is poured into the mould.
- Compact the moist soil into the mould in five layers of approximately equal mass, each layer being given 56 blows, with the help of 4.9 kg rammer, dropped from a height of 450 mm above the soil by using soil compactor. The blows must be distributed uniformly over the surface of each layer.
- After completion of the compaction operation, remove the extension collar and level carefully the top of the mould by means of straightedge.
- Now the CBR mould is taken and a filter paper is placed at the bottom. Inverse the compacted mould and take the weight of the sample along with mould.
- Surcharge weights, sufficient to produce an intensity of loading equal to the weight of the base material and pavement is placed on the specimen.

- To prevent upheaval of soil into the hole of the surcharge weights, 2.5 kg annular weight is placed on the soil surface prior to seating the penetration plunger after which the remainder of the surcharge weight is placed as shown in below the Figures.

Figure 5.1: CBR Moulds

- The mould is kept for soaking for a period of 96 hrs.
- After 96 hrs hours the mould is placed on the test plate. The plunger is to be seated under a load of 4 kg so that full contact is established between the surface of the specimen and the plunger.
- The stress and strain gauges are then set to zero. Load is applied to the penetration plunger so that the penetration is approximately 1.25 mm per minute.
- Readings of the load are taken at penetrations of 0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 4.0, 5.0, 7.5, 10.0 and 12.5 mm.
- The plunger is then raised and the mould detached from the loading equipment.
- The load penetration curve is plotted taking penetration value on x-axis and Load values on Y-axis as in below Table.
- Corresponding to the penetration value at which the CBR is desired, corrected load value is taken from the load-penetration curve and the CBR is calculated.

Table 5.1: CBR Results

LOAD PENETRATION TEST DATA						
GENERAL INFORMATION			**Penetration (mm)**	**Proving ring reading**	**Corrected load (Kg)**	**Proving ring reading**	**Corrected load (Kg)**
Type of compaction used: Dynamic			0.0	0	0.0	0	0.0
Period of soaking: 04 days			0.5	10	60.0	12	72.0
Wt.of surcharge used (kg): 5 Kgs			1.0	15	90.0	16	96.0
Proving ring capacity: 50 KN			1.5	19	113.9	20	119.9
Least count of proving dial guage:		5.997	2.0	22	131.9	24	143.9
			2.5	28	167.9	28	167.9
			3.0	34	203.9	33	197.9
			4.0	42	251.9	42	251.9
			5.0	50	299.9	51	305.8
			6.0	60	359.8	63	377.8
			7.5	79	473.8	81	485.8
			10.0	92	551.7	96	575.7
			12.5	110	659.7	112	671.7
Corrected unit load from graph(kg) for 2.5mm				170		165	

Corrected unit load from graph(kg) for 5.0mm	250	300
CBR at 2.5mm (%)	12.4	12.5
CBR at 5.0mm (%)	12.2	12.0
CBR Reported (%)	**12.4**	**12.5**

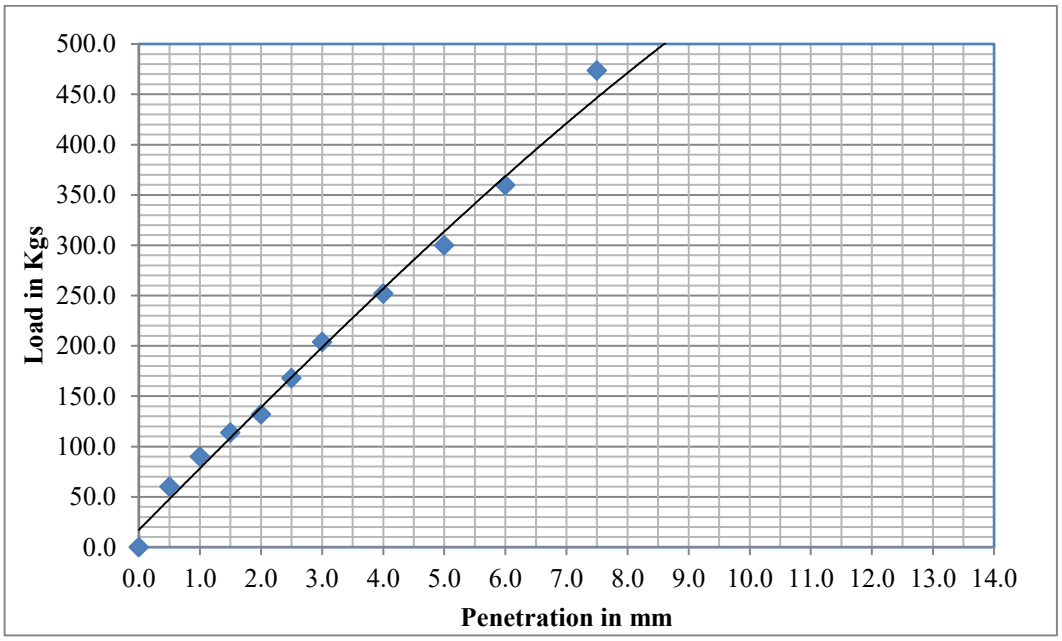


Figure 5.2: Sample Calculation Graph of CBR

- **Existed soil CBR values :** Collect existing / Embankment CBR Data as per SP 19 one sample for every one kilometer length. Compute 90th percentile CBR value for high volume roads and 80th percentile CBR value for low volume roads. For this study we considered 10km road. The test results of soaked CBR values given below;

Table 5.2 CBR Test results

Location (Chainage) (km)	4 days soaked CBR (%)
0	4.2
1	5.5
2	5.1
3	4.6
4	6.7
5	6.5
6	5.3
7	6.1
8	7.1
9	6.6

Table 5.3 Compute 90th percentile existing CBR value

Chainage (km)	4 days soaked CBR	Count	Decending order CBR	Percentile
0	4.2	10	4.2	100
1	5.5	9	4.6	90
2	5.1	8	5.1	80
3	4.6	7	5.3	70
4	6.7	6	5.5	60
5	6.5	5	6.1	50
6	5.3	4	6.5	40
7	6.1	3	6.6	30
8	7.1	2	6.7	20
9	6.6	1	7.1	10

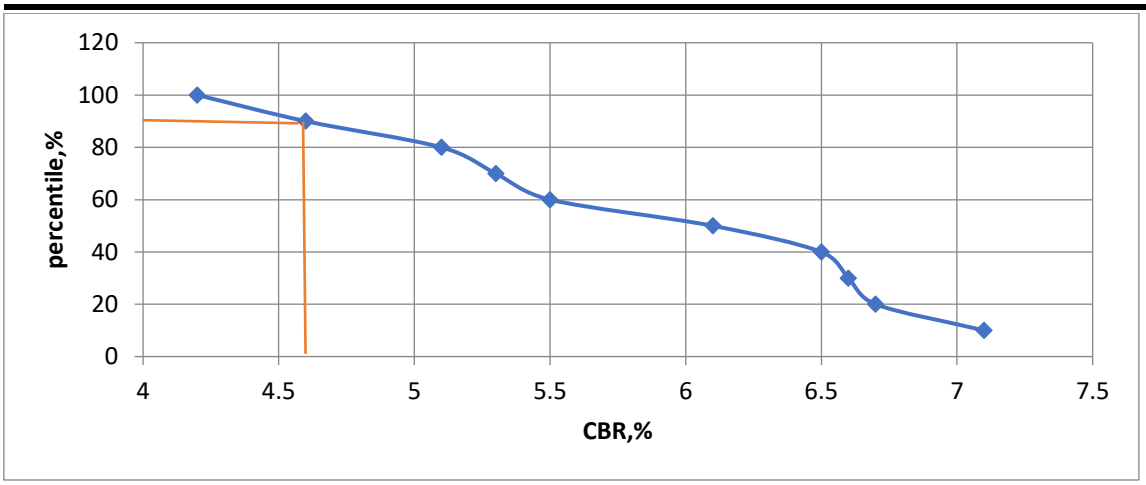


Figure 5.3: 90th percentile CBR value

Result: 90th percentile CBR value found as **4.6% &** CBR value of borrow soil is 8.5 %

5.2 TRAFFIC VOLUME

5.2.1 Commercial Vehicle Per Day:

Based on traffic volume count survey carried out on project corridor as per IRC SP 19-2001 for 7days 24 hour. Average of 7days traffic consider as average daily traffic (ADT). Only commericial vehicles (having the laden weight of 3 Tonnes or more) count to be considered.

Table 5.4: Commercial Vehicle Per Day

DAY	CVPD (Two ways)
Monday	2502
Tuesday	2365
Wednesday	2743
Thursday	2100
Friday	2268
Saturday	2598
Sunday	2643
Average	2459.85 ≈ 2460

Result: Initial CVPD based on traffic volume survey = 2460 (Two ways)

CHAPTER 6

IITPAVE SOFTWARE

IITPAVE software is used to calculate the Actual Horizontal Tensile Strain in Bituminous layer and Actual Vertical Compressive Strain on sub-grade.

The actual strains are computed using various trial pavement thickness combinations. The tyre pressure used in the analysis is 0.56 MPa. Standard axle used is dual type and single tyre load of 20,000 N. The Poisson's ratio of bituminous layer is taken as 0.35. The E values and Poisson's ratio obtained from above are given as inputs to the software and strains are calculated.

6.1 LAUNCHING THE SOFTWARE

1. From the Home screen user can manually give input through input window by clicking on 'Design New Pavement Section'. User can also give input through properly formatted input file by clicking on 'Edit Existing File' option then browsing and opening the input file.
2. Next an input window will come. All the inputs required have to be given through that input window.

6.2 INPUTS TO THE SOFTWARE

1. First, number of layers to be selected from drop down menu to fix up input boxes for different layer.
2. Next, Elastic modulus (E) values of the various layers in MPa, Poisson's ratio and thickness of the layers in mm excluding the subgrade thickness are to be provided.
3. Single wheel load and the tyre pressure are to be provided (tyre pressure of 0.56 MPa has been used for calibration of the fatigue equation and the same pressure can be used for stress analysis. Change of pressure even up to 0.80 MPa has a small effect upon stress values in lower layers.)
4. Then the number of points for stress computations is to be given through the drop down menu for Analysis points.
5. Then corresponding to different points, the values of depth Z in mm and the corresponding value of radial distances from wheel centre (r) in mm are to be provided.
6. Provide whether analysis is for single wheel load or double wheel load by clicking 1 or 2 and 2 will be the most common case.

CHAPTER 7

DESIGN AND ANALYSIS OF PAVEMENT

7.1 GENERAL

The pavement design basically aims at determining the total thickness of the pavement structure as well as thickness of individual structural components. Pavement is the most significant component of a road and therefore its design strengths must be assured to support the projected traffic loading throughout the design period.

The pavement design is carried out for flexible by using IRC and IIT PAVE methods. The overall thickness has been worked out, the results were compared, and the optimized solution based on characteristics of existing materials, best engineering judgment and environmental conditions has been adopted. In this chapter we shall discuss the design of Pavement by using the IRC 37: 2018 and IIT PAVE software.

7.2 IRC GUIDELINES

The guidelines on design of flexible pavement were first brought out in 1970, which were based on California Bearing Ratio (CBR) of subgrade and traffic in terms of number of commercial vehicles. The guidelines were revised again in 2001 when pavements were required to be designed for traffic as high as 150 msa. The revised guidelines used a semi-mechanistic approach based on the results of the MORTH's research scheme R-56 implemented at IIT Kharagpur.

The flexible pavement has low flexural strength and hence layers reflect the deformation of the lower layers/ subgrade on to the surface layer after the withdrawal of wheel load. To control the deflections in the subgrade so that no permanent deflections results the pavement thickness is so designed that the stresses on the subgrade soil are kept within its bearing power. Loading of bituminous pavement requires the stiffest layers to be placed at the surface with successive weaker layers down to subgrade.

The traffic pattern has changed so has the technology. The volume of tandem, tridem and multi-axle vehicles has increased manifold and heavier axle loads are common. Conventional construction material like aggregates is becoming progressively scarce on account of environmental concerns as well as legal restrictions on quarrying while the construction activity has expanded phenomenally. This has shifted focus from large scale use

of conventional aggregates to use of local, recycled and engineered marginal aggregates in construction.

Accordingly, the revision of IRC 37 - 2018 incorporates some of the new and alternate materials in the current design practices. A designer can use his sound engineering judgment consistent with local environment using a semi-mechanistic approach for design of pavements.

7.3 DESIGN OF PAVEMENT (IRC 37: 2018 AND IIT PAVE)

As discussed earlier the design of flexible pavements is done by using the guidelines of IRC 37: 2012 and applying the results to the IIT PAVE software.

7.3.1 IRC 37:2018

Indian roads congress has specified the design procedures for flexible pavements based on CBR values. The Pavement designs given in the previous edition IRC:37-1984 were applicable to design traffic up to only 30 million standard axles (msa). The earlier code is empirical in nature which has limitations regarding applicability and extrapolation. These guidelines follow analytical designs and developed new set of designs up to 150 msa.

These guidelines will apply to design of flexible pavements for Expressway, National Highways, State Highways, Major District Roads, and other categories of roads. Flexible pavements are considered to include the pavements which have bituminous surfacing and granular base and sub-base courses conforming to IRC/ MOST standards. These guidelines apply to new pavements.

In M-E pavement design approach, the pavement is idealized as a layered structure (generally assumed as elastic for simplicity in analysis) consisting of three to four horizontal layers made up of bituminous surfacing, base, sub-base and the subgrade. Each layer is characterized by its elastic modulus, Poisson's ratio and the thickness. Fatigue, rutting and low temperature cracking are generally considered as the important modes of failure of a bituminous pavement structure.

Indian Roads Congress Method is based on an empirical method where the thickness value of a pavement used was read from the CBR value of the sub-grade. From the design chart the total pavement thickness could be read for a given CBR value and cumulative standard axle load.

In our design the pavement is divided into four layers i.e Bituminous course, DBM layer, WMM layer and Subgrade. So we need to analyse the properties of all these layers.

The design steps for pavement given below;

1. Compute subgrade effective resilient modulus (M_{RS})
2. Compute design traffic in terms of MSA (N_{DES})
3. Select layer combination
4. Compute permissiable horizontal & vertical strain
5. Compute actual strain as per IITPAVE.(try combinations of different thicknesses)
6. Selection of Economic pavement composition

7.3.1.1 Resilient Modulus and Poisson Ratio of Subgrade

The subgrade is the top 500 mm of the embankment immediately below the bottom of the pavement, and is made up of in-situ material, selected soil, or stabilized soil that forms the foundation of a pavement. It should be well compacted to limit the scope of rutting in pavement due to additional densification during the service life of pavement.

The behaviour of the subgrade is essentially elastic under the transient traffic loading with negligible permanent deformation in a single pass. Resilient modulus is the measure of its elastic behaviour determined from recoverable deformation in the laboratory tests. The modulus is an important parameter for design and the performance of a pavement.

The relation between modulus and the effective CBR in IRC is given as (Annexure 1)

$$M_{RS}\ (MPa) = 10 * CBR \qquad \text{for } CBR <= 5$$

$$= 17.6 * (CBR)^{0.64} \qquad \text{for } CBR > 5$$

M_{RS} = modulus of subgrade soil.

From the **5.3.1.5** in chapter 5 : 90th percentile CBR value found as **4.6% &** CBR value of borrow soil is **8.5 %.**

Existing soil M_{RS} (Mpa) = 10 * (CBR) = 10 * 4.6 = 46 Mpa

Borrow soil M_{RS} (Mpa) = $17.6 * (CBR)^{0.64} = 17.6 * (8.5)^{0.64}$ = 69.237 Mpa

Wheel load 40000Newtons & tyre pressure 0.56

Then submit these inputs in IITPAVE and the strains are obtained as shown in Figure 7.3(a & b).

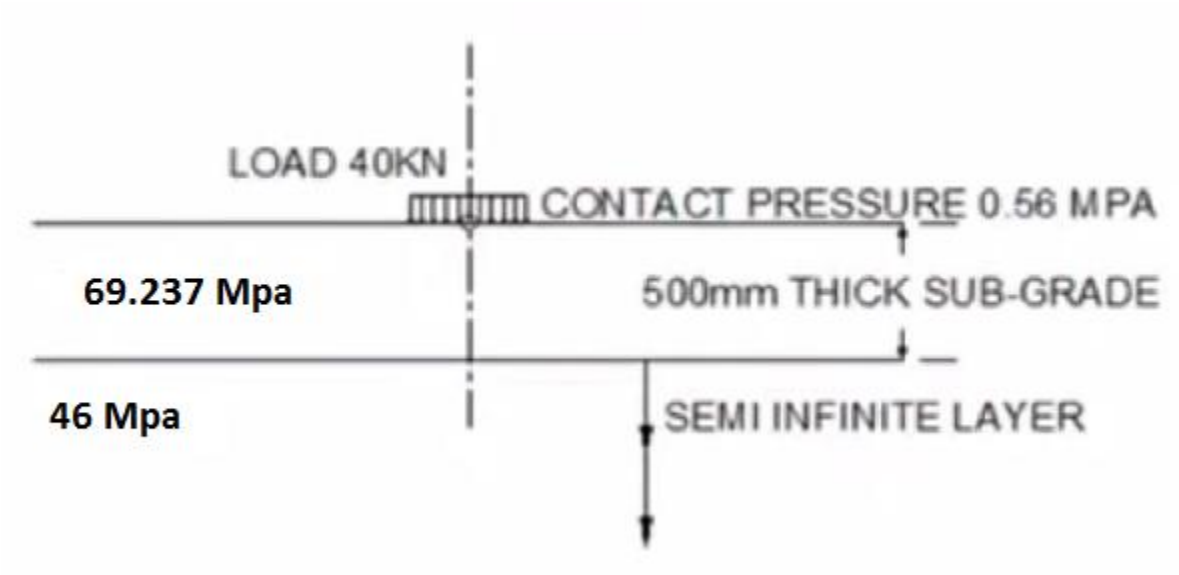


Figure 7.1 resilient modulus values with respect to the layer

LAYER	POISON RATION
BITUMINOUS LAYER	0.35
WMM	0.35
CTB	0.25
GSB	0.35
CTSB	0.25
AGGREGATE INTER LAYER	0.35
RAP BASE	0.35
SUB-GRADE	0.35

Figure 7.2 Poisson ratio

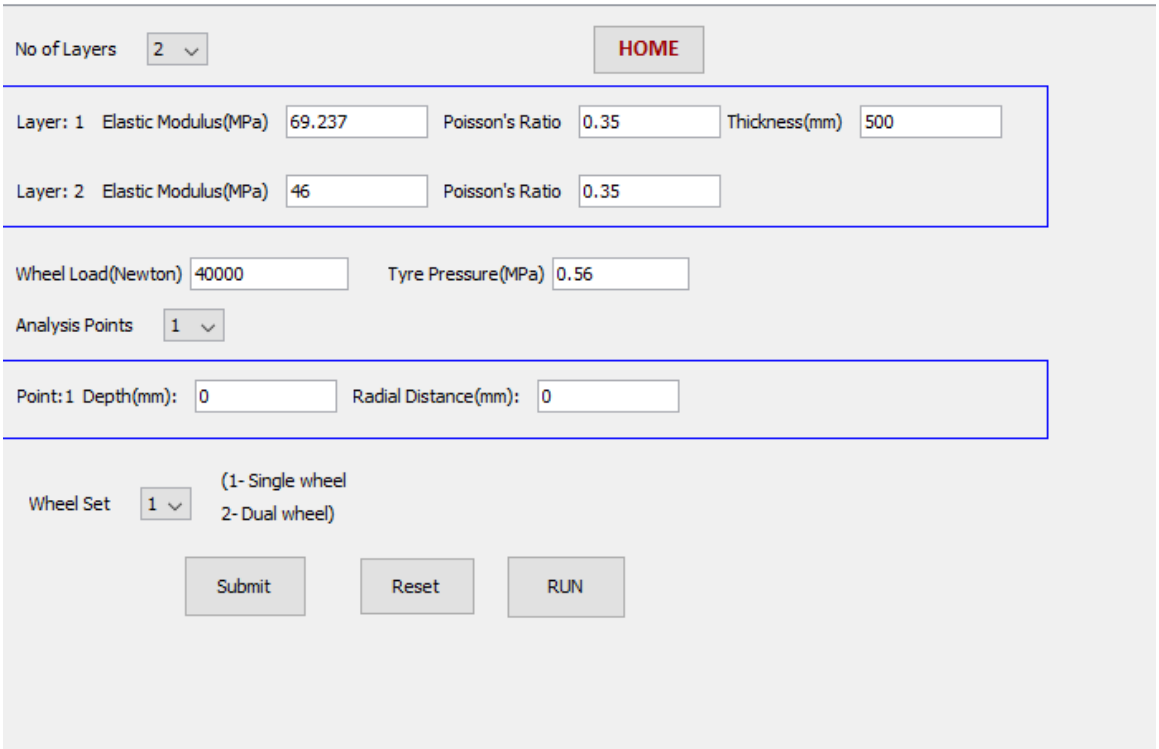


Figure 7.3(a) Input file

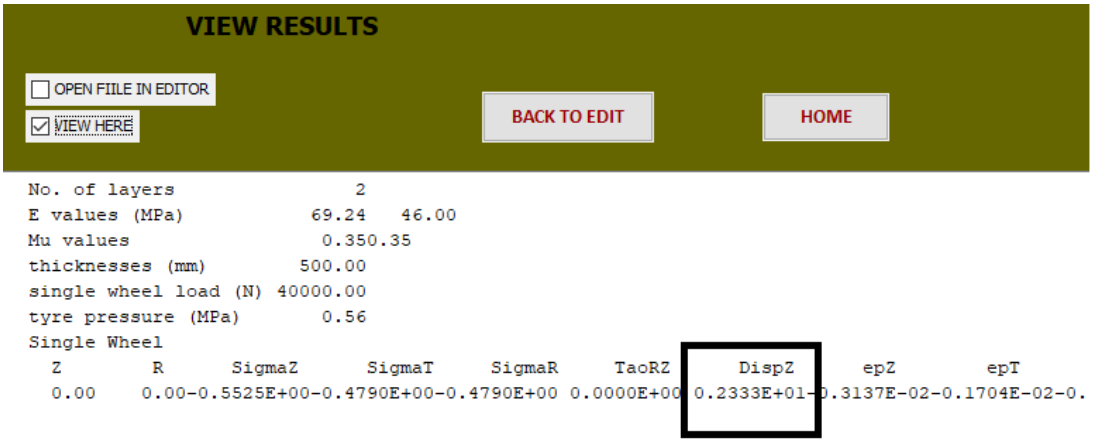


Figure 7.3(b)Computed deflection

$M_{RS} = [2(1 - \mu^2)\ pa\ ] / \delta$

Where,

p = contact pressure = 0.56 MPa

a = radius of circular contact area, which can be calculated using the load applied (40,000 N) and the contact pressure 'p' (0.56 MPa) = 150.8 mm

μ = Poisson's ratio

$M_{RS} = [2(1 - \mu^2)\ pa\ ] / \delta = [2(1-0.35^2\ ) \times 0.56 \times 150.8] / 2.333 = 63.526$ Mpa

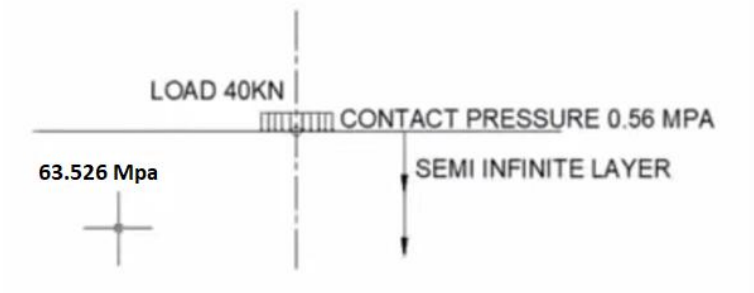


Figure 7.4 resilient modulus for combined borrow soil & existed soil

7.3.1.2 Design Traffic

The design traffic, in terms of the cumulative number of standard axles to be carried during the design period of the road, should be estimated using by below equation;

$$N_{Des} = [(365\ ((1+r)^n - 1) / r)) * A * D * F]$$

Where,

N_{Des} = cumulative number of standard axles to be catered for during the design period of 'n' years

A = initial traffic (commercial vehicles per day) in the year of completion of construction (directional traffic volume to be considered for divided carriageways where as for other categories of the carriageway, two-way traffic volume may be considered for applying the lateral distribution factors)

D = lateral distribution factor (as explained in para 4.5 IS 37-2018)

F = vehicle damage factor (VDF)

n = design period, in years

r = annual growth rate of commercial vehicles in decimal (e.g., for 5 per cent annual growth rate, r = 0.05). Variation of the rate of growth over different periods of the design period, if available, may be considered for estimating the design traffic

The traffic in the year of completion of construction may be estimated by using below equation;

$$A = P(1+r)^x$$

Where,

P = number of commercial vehicles per day as per last count.

x = number of years between the last count and the year of completion of construction.

Commercial vehicles per day (P): Based on traffic count survey carried out on projrct corridor as per IRC SP 19 2001 for 7days 24 hours. Average of 7days traffic consider as average daily traffic (ADT). Only commercial vehicles count to be considered.

Annual growth rate (r): Annual growth rate can be derived as per IRC 108 2015 by using past NSDP, GDP, PCI, Vehicle registration, Fuel consumption and population data. A minimum annual growth rate of 5% should be used for commercial vehicles for estimating the design traffic.

Design period (n) : It recommended that a design period of 20years may be adopted for the structural design of pavements for National highways, state highways and urban roads. For other categories of roads, a design period of 15 years is recommended. Pavement for very high density corridors (more than 300msa) and expressways shall preferably be designed as long life pavements.

Lateral distribution factor(D) : Lateral distribution of commercial traffic on the carriageway is required for estimated the design traffic (Equivalent standard axle load application). To be considered for the structural design of pavement.

HIGHWAY	DISTRIBUTION FACTOR
SINGLE LANE ROAD	100 PERCENT COMMERCIAL VEHICLE OF BOTH DIRECTIONS.
INTERMEDIATE LANE 5.5 m WIDTH	75 PERCENT COMMERCIAL VEHICLE OF BOTH DIRECTIONS.
TWO LANE TWO WAY	50 PERCENT COMMERCIAL VEHICLE OF BOTH DIRECTIONS.
FOUR LANE SINGLE CARRIAGEWAY	40 PERCENT COMMERCIAL VEHICLE OF BOTH DIRECTIONS.
DUAL TWO LANE CARRIAGEWAY	75 PERCENT COMMERCIAL VEHICLE OF EACH DIRECTION.
DUAL THREE LANE CARRIAGEWAY	60 PERCENT COMMERCIAL VEHICLE OF EACH DIRECTION.
DUAL FOUR LANE CARRIAGEWAY	45 PERCENT COMMERCIAL VEHICLE OF EACH DIRECTION.

Figure 7.5 : Lane distribution factor

Data for design:

- Dual Three lane carriageway : Distribution factor = D = 60/100 = 0.6
- Initial CVPD based on traffic volume survey = 2460 CVPD (two ways) = P
- Traffic growth rate = r = 5%
- Vehicle damage factor = F = 5 (same for both direction)
- Construction period =x = 1 year
- Design period = n = 20years

Derive Initial traffic in year of construction:

$A = P(1+r)^x = 2460\ (1+0.05)^1 = 2583$ CVPD (two ways) ≈ 2584 CVPD = 1292(assume 50 % in each side)

Derive MSA:

$N_{DES} = [(365\ ((1+r)^n - 1) / r)) \times A \times D \times F]$

$= [(365\ (1+0.05)^{20} - 1) / 0.05) \times 1292 \times 0.6 \times 5] = 46.7797$ MSA = Say **47MSA**

7.3.1.3 Layer Selection

Bituminous Layer : The surfacing consists of a wearing course or a binder course plus wearing course. The most commonly used wearing courses are surface dressing, open graded premix carpet, mix seal surfacing, semi-dense bituminous concrete and bituminous concrete. For binder course, MORTH specifies, it is desirable to use bituminous macadam (BM) for traffic upto to 5 msa and dense bituminous macadam (DBM) for traffic more than 5 msa. The modulus for the bituminous material are obtained from the Table 9.2 in IRC 37: 2018

Mix type	Average Annual Pavement Temperature °C				
	20	25	30	35	40
BC and DBM for VG10 bitumen	2300	2000	1450	1000	800
BC and DBM for VG30 bitumen	3500	3000	2500	2000	1250
BC and DBM for VG40 bitumen	6000	5000	4000	3000	2000
BC with Modified Bitumen (IRC: SP: 53)	5700	3800	2400	1600	1300
BM with VG10 bitumen	500 MPa at 35°C				
BM with VG30 bitumen	700 MPa at 35°C				
RAP treated with 4 per cent bitumen emulsion/ foamed bitumen with 2-2.5 per cent residual bitumen and 1.0 per cent cementitious material.	800 MPa at 35°C				

Figure 7.6: Indicative values of resilient modulus (MPa) of bituminous mixes

Note: For the purpose of the design

a. Resilient modulus measured at 35°C temperature as per ASTM 4123 shall be adopted. For snowbound areas resilient modulus shall be measured at 20ºC

b. The same indicative maximum modulus values are recommended for BC (surface course) as well as DBM (binder/base course) with unmodified binders

c. The resilient modulus values for surfacing courses with modified bitumen shall be taken to be same as the resilient modulus values indicated for DBM

In this study we designed pavement by adopting VG40 bitumen for both DBM & BC : the resilent modulus is **3000Mpa**

LAYERS		MIN. THICKNESS AS PER IRC 37 & MoRTH
BITUMINOUS	BC/SDBC/GGRB	30 MM
	SMA AS WEARING COURSE	40 MM
	SMA AS BASE COURSE	45 MM
	DBM/BM	50 MM
GRANULAR	WMM/WBM/CRM/RC/GSB	150 MM
	CTB	100 MM
	CTSB	200 MM
	SUB-GRADE	500 MM

Figure 7.7: Min Layer thickness as per IRC 37

Existence soil soaked CBR value of 4.6% & 47 MSA

Figure 12.1 Catalogue for pavement with bituminous surface course with granular base and sub-base - Effective CBR 5% (Plate-1)

GSB = 200mm

WMM = 250mm

BASE / BINDER COURSE = 140mm

SURFACE COURSE = 40mm

SUBGRADE = 500mm

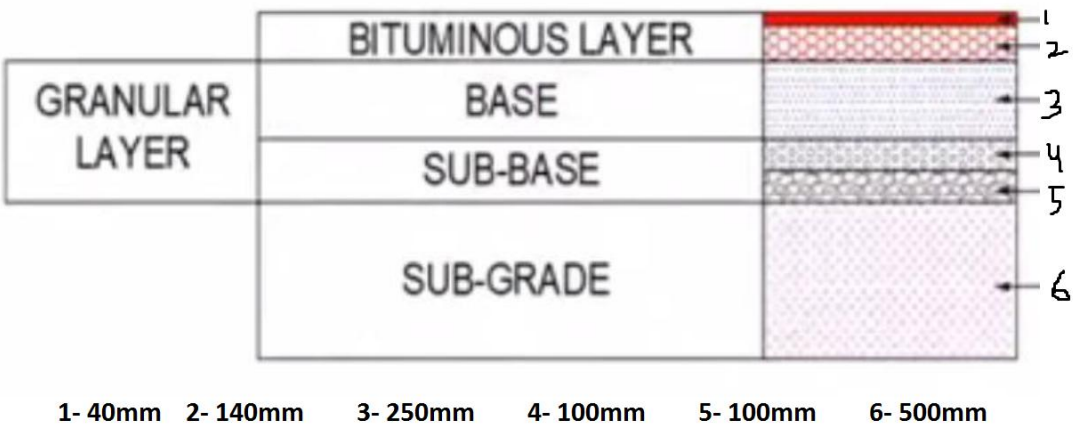


Figure 7.8: Adopted layer thickness

7.3.1.4 Failure Criteria and Strain Calculation

The flexible pavements has been modeled as a three layer structure and stresses and strains at critical locations have been computed using the linear elastic model. To give proper consideration to the aspects of performance, the following three types of pavement distress resulting from repeated (cyclic) application of traffic loads are considered

- Vertical compressive strain at the top of the sub-grade which can cause sub-grade deformation resulting in permanent deformation at the pavement surface.
- Horizontal tensile strain or stress at the bottom of the bituminous layer which can cause fracture of the bituminous layer.
- Pavement deformation within the bituminous layer.

While the permanent deformation within the bituminous layer can be controlled by meeting the mix design requirements, thickness of granular and bituminous layers are selected using the analytical design approach so that strains at the critical points are within the allowable limits.

The figure 7.9 shows the cross section of a bituminous pavement with granular base and subbase. It is considered as a three layer elastic structure consisting of bituminous surfacing, granular base and subbase and the subgrade. The granular layers are treated as a single layer. Strain and stresses are required only for three layer elastic system. The critical are located at the bottom of bituminous layer and at top of subgrade.

- **Fatigue Criteria:**

Bituminous surfacings of pavements display flexural fatigue cracking if the tensile strain at the bottom of the bituminous layer is beyond certain limit. The relation between the fatigue life of the pavement and the tensile strain in the bottom of the bituminous layer was obtained as

$$N_f = 1.6064*C*10^{-04}\,[1/\varepsilon_t]^{3.89}*\,[1/M_{Rm}]^{0.854} \quad \text{(for 80 \% reliability)}$$

$$N_f = 0.5161*C*10^{-04}\,[1/\varepsilon_t]^{3.89}*\,[1/M_{Rm}]^{0.854} \quad \text{(for 90 \% reliability)}$$

Where

$$C = 10^{M}, \quad \text{and} \quad M = 4.84\left(\frac{Vbe}{Va+Vbe} - 0.69\right)$$

V_a = per cent volume of air void in the mix used in the bottom bituminous layer

V_{be} = per cent volume of effective bitumen in the mix used in the bottom bituminous layer

N_f = fatigue life of bituminous layer (cumulative equivalent number of 80 kN standard axle loads that can be served by the pavement before the critical cracked area of 20 % or more of paved surface area occurs)

ε_t = maximum horizontal tensile strain at the bottom of the bottom bituminous layer (DBM) calculated using linear elastic layered theory by applying standard axle load at the surface of the selected pavement system

M_{Rm} = resilient modulus (MPa) of the bituminous mix used in the bottom bituminous layer, selected as per the recommendations made in these guidelines.

Reliability: 90% Reliability for all important roads such as expressways , national highways, state highways, urban road and for other categories of road for design traffic 20MSA or more. 80% reliability for design traffic less than 20msa.

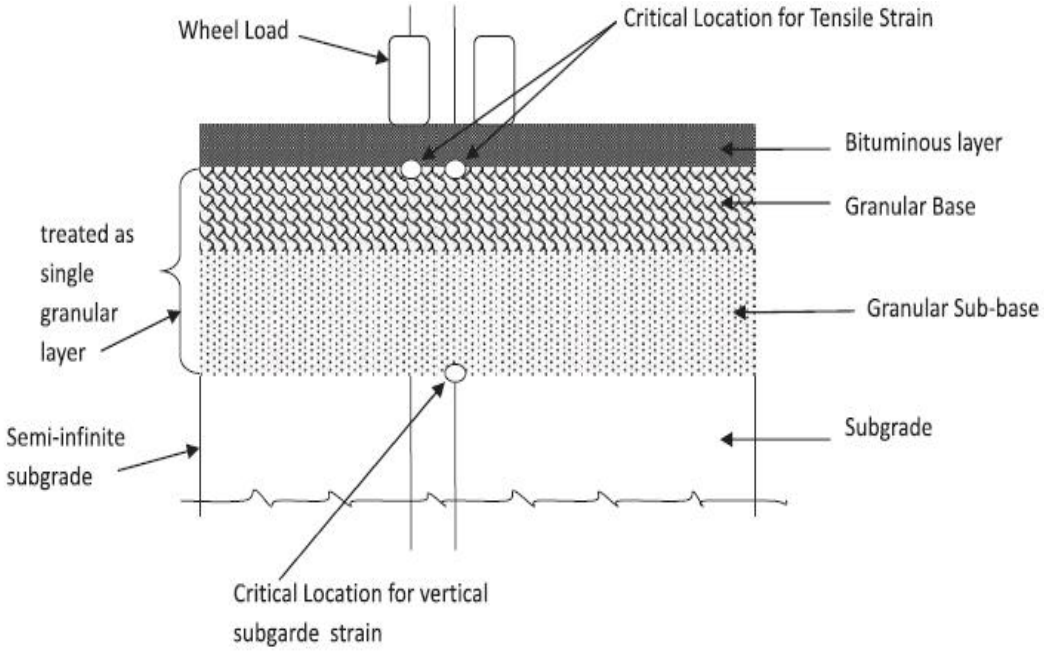


Figure 7.9: Critical Location of Strains

- **Rutting Criteria**

The allowable number of load repetitions to control permanent deformation can be expressed as

$$N_R = 4.1656 \times 10^{-08} [1/\varepsilon_v]^{4.5337} \quad \text{(for 80 \% reliability)}$$

$$N_R = 1.4100 \times 10^{-08} [1/\varepsilon_v]^{4.5337} \quad \text{(for 90 \% reliability)}$$

Where

N_R = subgrade rutting life (cumulative equivalent number of 80 kN standard axle loads that can be served by the pavement before the critical rut depth of 20 mm or more occurs)

ε_v = vertical compressive strain at the top of the subgrade calculated using linear elastic layered theory by applying standard axle load at the surface of the selected pavement system

The developed critical initial strain values for each of the failure modes are computed from suitable pavement analysis routine. The allowable strain values are obtained from the input parameters (related to pavement material, environment, traffic and design life). Finally, the process of pavement design involves adjusting and subsequently selecting the appropriate thickness values of various layers so that the critical strain parameters are within the allowable limits. The design thickness values, for a given set of input parameters can be estimated from ready-made pavement design charts available in various guidelines.

PERMISSIBLE STRAIN:

a) Vertical strain (90% reliability) : rutting

$N_R = 1.41 \text{ X } 10^{-8} [1/ \varepsilon_v]^{4.5337}$

$47 \text{ x } 10^6 = 1.41 \text{ x } 10^{-8} \text{ x } [1/ \varepsilon_v]^{4.5337}$

$\varepsilon_v = 0.0003768$

b) Horizontal strain (90% reliability): fatigue

$N_F = 0.561 \text{ X C X } 10^{-4} \text{ X } [1/ \varepsilon_t]^{3.89} \text{ x } [1/M_{RM}]^{0.854}$

$C = 10^M$ & $M = 4.84 [V_{be} / (V_{be} + V_a)]^{4.5337}$

Use, V_{be} =11.5 (Actual V_{be} : 11 to 13) & V_a = 3.5 (Actual V_a : not less than 3%)

M = 0.37 , C = 2.35

M_{RM} = 3000 Mpa (Figure 7.6)

$\varepsilon_t = 0.0001849095$

7.3.1.5 Modulus for each layer *:*

sub-grade modulus = 48.736 Mpa (From figure 7.4)

bitumen layer modulus = 3000 Mpa (From figure 7.6)

Granular DBM layer:

Sub-base materials comprise natural sand, gravel, laterite, brick metal, crushed stone or combinations there of meeting the prescribed grading and physical requirements. The sub-base material should have a minimum CBR of 20 % and 30 % for traffic upto 2 msa and traffic exceeding 2 msa respectively. Sub-base usually consist of granular or WBM and the thickness should not be less than 150 mm for design traffic less than 10 msa and 200 mm for design traffic of 1.0 msa and above. The modulus is found out by

Granular layer modulus = $M_{GRAN} = 0.2 (h)^{0.45} \text{ x } M_{support}$

h = Thickness of the DBM layer

M_{GRAN} = 0.2 $(250+200)^{0.45}$ x 48.736 = 152.344 Mpa

Granular layer modulus = 152.344 Mpa

7.3.1.6 Actual strain by IITPAVE

Inputs:

Wheel load = 20000 N

Tyre Pressure = 0.56 MPa

Poissons's Ratio = 0.35

Radial distance = 155mm

Strain Comparision:

1. Actual horizontal strain < = Permissible horizontal strain

2. Actual vertical strain < = Permissible vertical strain

<u>Trail – 1:</u>

SURFACE COURSE = 30mm, BASE / BINDER COURSE = 100mm, WMM = 250mm, GSB = 200mm, SUBGRADE = 500mm

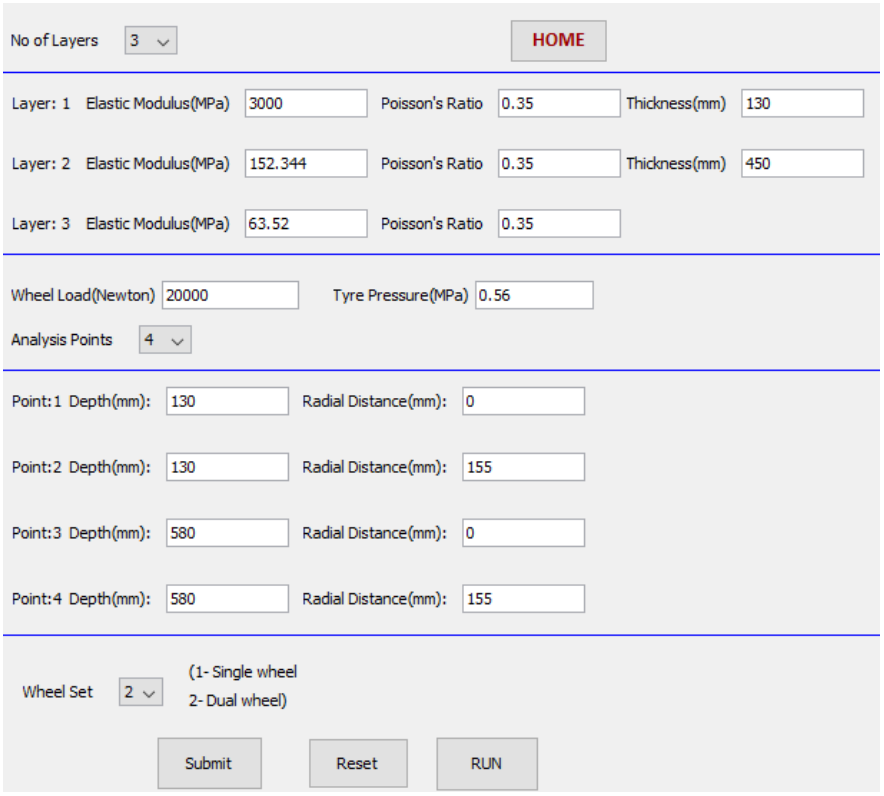
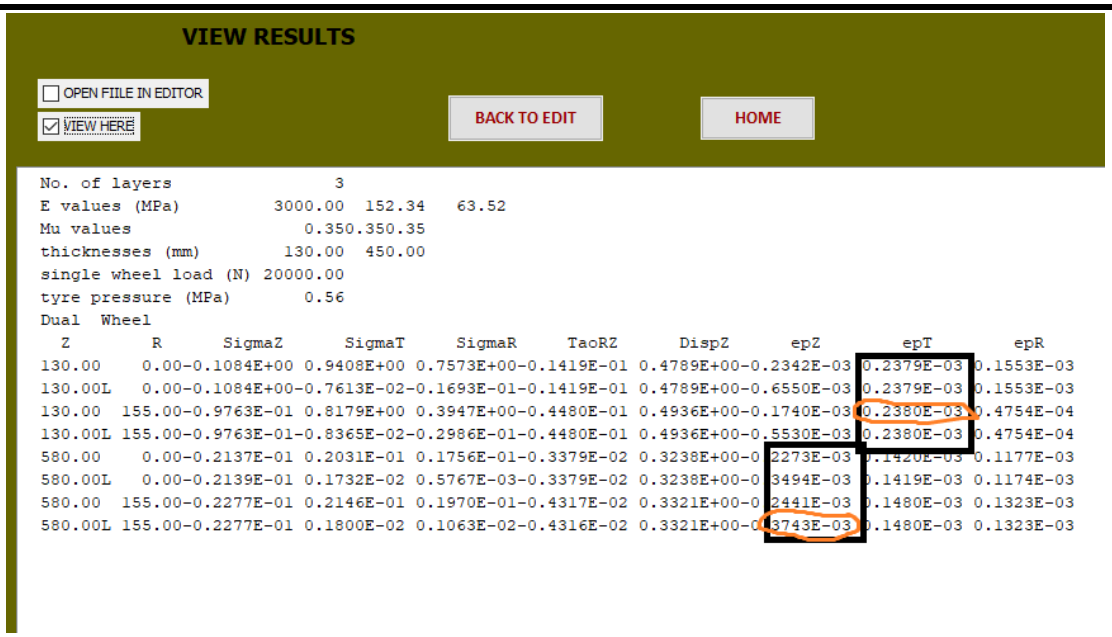


Figure 6.10: Trail 1 Input



VIEW RESULTS

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```
No. of layers                3
E values (MPa)          3000.00  152.34   63.52
Mu values                  0.350.350.35
thicknesses (mm)         130.00  450.00
single wheel load (N) 20000.00
tyre pressure (MPa)        0.56
Dual  Wheel
  Z        R      SigmaZ      SigmaT      SigmaR      TaoRZ       DispZ       epZ         epT         epR
130.00     0.00-0.1084E+00 0.9408E+00 0.7573E+00-0.1419E-01 0.4789E+00-0.2342E-03 0.2379E-03 0.1553E-03
130.00L    0.00-0.1084E+00-0.7613E-02-0.1693E-01-0.1419E-01 0.4789E+00-0.6550E-03 0.2379E-03 0.1553E-03
130.00   155.00-0.9763E-01 0.8179E+00 0.3947E+00-0.4480E-01 0.4936E+00-0.1740E-03 0.2380E-03 0.4754E-04
130.00L  155.00-0.9763E-01-0.8365E-02-0.2986E-01-0.4480E-01 0.4936E+00-0.5530E-03 0.2380E-03 0.4754E-04
580.00     0.00-0.2137E-01 0.2031E-01 0.1756E-01-0.3379E-02 0.3238E+00-0.2273E-03 0.1420E-03 0.1177E-03
580.00L    0.00-0.2139E-01 0.1732E-02 0.5767E-03-0.3379E-02 0.3238E+00-0.3494E-03 0.1419E-03 0.1174E-03
580.00   155.00-0.2277E-01 0.2146E-01 0.1970E-01-0.4317E-02 0.3321E+00-0.2441E-03 0.1480E-03 0.1323E-03
580.00L  155.00-0.2277E-01 0.1800E-02 0.1063E-02-0.4316E-02 0.3321E+00-0.3743E-03 0.1480E-03 0.1323E-03
```

Figure 6.11: Trail 1 Output

Strain Comparision:

Permissible horizontal strain = 0.0001849095

Actual horizontal strain = 0.0002380

Permissible vertical strain = 0.0003768

Actual vertical strain = 0.0003743

Here, 1. Actual horizontal strain > Permissible horizontal strain

2. Actual vertical strain < Permissible vertical strain

Hence Pavement is unsafe.

Trail – 2:

SURFACE COURSE = 30mm, BASE / BINDER COURSE = 120mm, WMM = 250mm, GSB = 200mm, SUBGRADE = 500mm

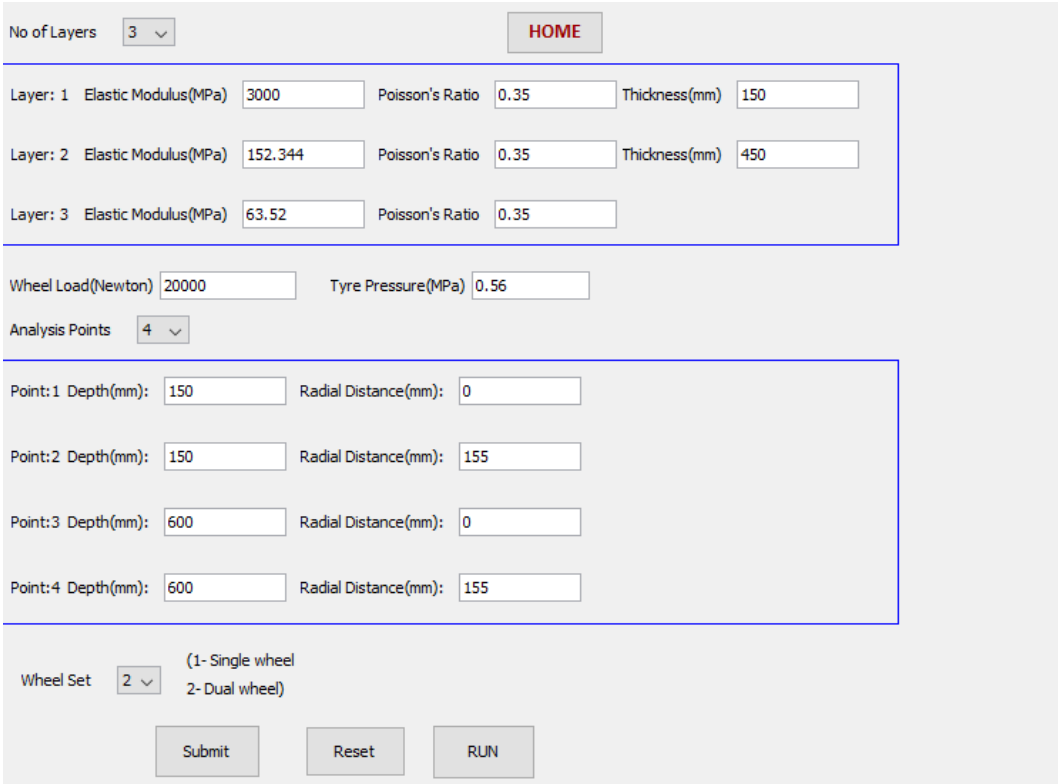


Figure 6.12: Trail 2 Input

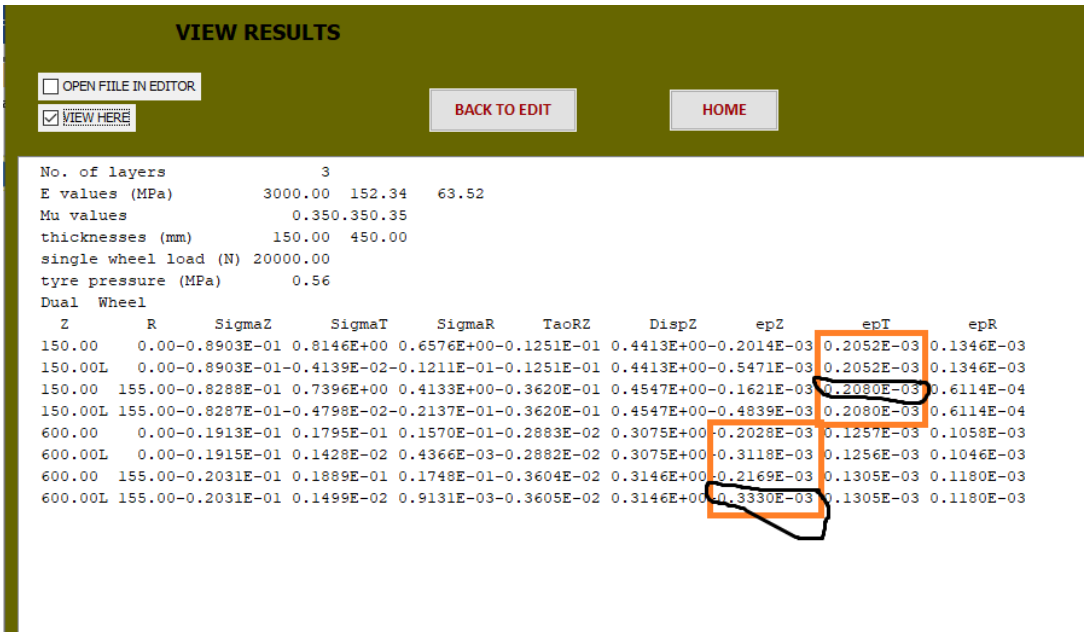


Figure 6.13: Trail 2 Output

Strain Comparision:

Permissible horizontal strain = 0.0001849095

Actual horizontal strain = 0.0002080

Permissible vertical strain = 0.0003768

Actual vertical strain = 0.000333

Here, 1. Actual horizontal strain > Permissible horizontal strain

2. Actual vertical strain < Permissible vertical strain

Hence Pavement is Un-Safe.

Trail – 3:

SURFACE COURSE = 30mm, BASE / BINDER COURSE = 150mm, WMM = 250mm, GSB = 200mm, SUBGRADE = 500mm

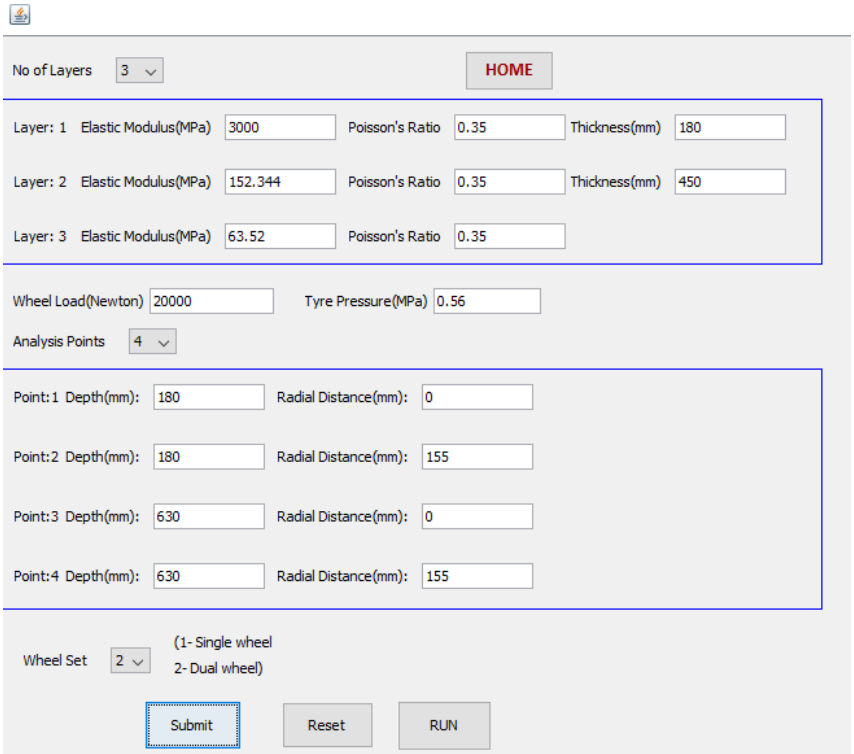
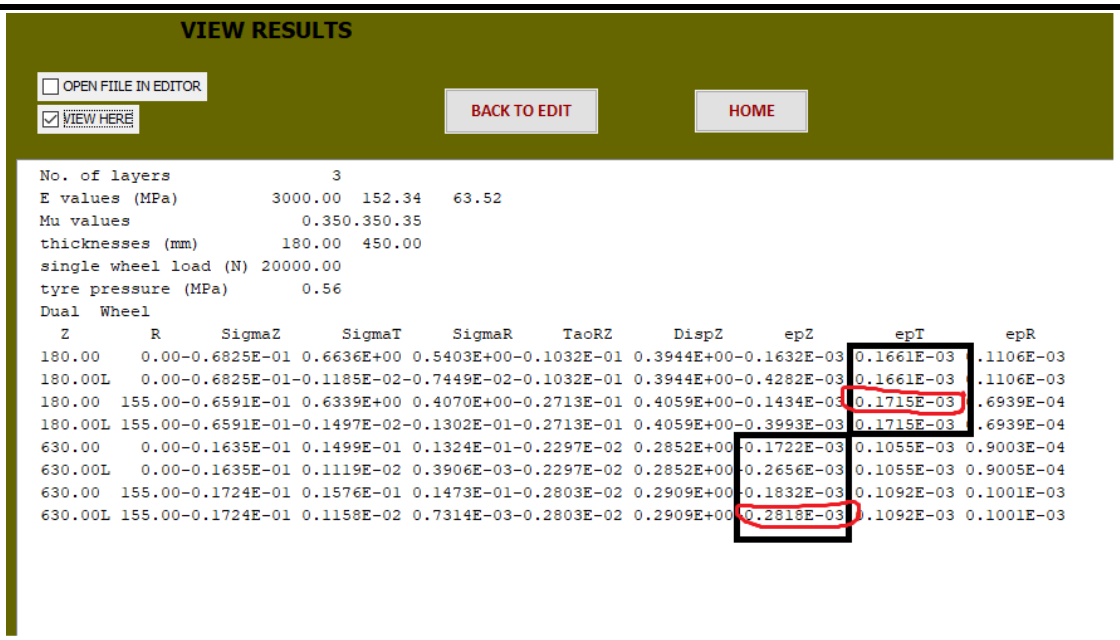


Figure 6.14: Trail 3 Input



VIEW RESULTS

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No. of layers 3
E values (MPa) 3000.00 152.34 63.52
Mu values 0.350.350.35
thicknesses (mm) 180.00 450.00
single wheel load (N) 20000.00
tyre pressure (MPa) 0.56
Dual Wheel

Z	R	SigmaZ	SigmaT	SigmaR	TaoRZ	DispZ	epZ	epT	epR
180.00	0.00	-0.6825E-01	0.6636E+00	0.5403E+00	-0.1032E-01	0.3944E+00	-0.1632E-03	0.1661E-03	.1106E-03
180.00L	0.00	-0.6825E-01	-0.1185E-02	-0.7449E-02	-0.1032E-01	0.3944E+00	-0.4282E-03	0.1661E-03	.1106E-03
180.00	155.00	-0.6591E-01	0.6339E+00	0.4070E+00	-0.2713E-01	0.4059E+00	-0.1434E-03	0.1715E-03	.6939E-04
180.00L	155.00	-0.6591E-01	-0.1497E-02	-0.1302E-01	-0.2713E-01	0.4059E+00	-0.3993E-03	0.1715E-03	.6939E-04
630.00	0.00	-0.1635E-01	0.1499E-01	0.1324E-01	-0.2297E-02	0.2852E+00	-0.1722E-03	0.1055E-03	0.9003E-04
630.00L	0.00	-0.1635E-01	0.1119E-02	0.3906E-03	-0.2297E-02	0.2852E+00	-0.2656E-03	0.1055E-03	0.9005E-04
630.00	155.00	-0.1724E-01	0.1576E-01	0.1473E-01	-0.2803E-02	0.2909E+00	-0.1832E-03	0.1092E-03	0.1001E-03
630.00L	155.00	-0.1724E-01	0.1158E-02	0.7314E-03	-0.2803E-02	0.2909E+00	-0.2818E-03	.1092E-03	0.1001E-03

Figure 6.15: Trail 3 Output

Strain Comparision:

Permissible horizontal strain = 0.0001849095

Actual horizontal strain = 0.0001715

Permissible vertical strain = 0.0003768

Actual vertical strain = 0.0002818

Here, 1. Actual horizontal strain < Permissible horizontal strain

2. Actual vertical strain < Permissible vertical strain

Hence Pavement is Safe.

7.3.1.7 Selection of Economic pavement composition

With try of various thickness combination one can compare and compute the economical option for pavement structure.

Table.7.1 Economic pavement composition

Trail	Surface course (mm)	Binder Course (mm)	WMM(mm)	GSB (mm)	Subgrade (mm)	Permissible strain		Actual strain		Safe / unsafe
						Horizontal	Vertical	Horizontal	Vertical	
1	30	100	250	200	500	0.0001849095	0.0003768	0.0002404	0.0004355	unsafe
2	30	120	250	200	500			0.0002080	0.000333	unsafe
3	30	150	250	200	500			0.0001715	0.0002818	**safe**
3	30	150	200	200	500			0.0001731	0.0003101	**safe**
4	30	120	200	200	500			0.0002099	0.0003699	unsafe
5	30	130	200	200	500			0.0001966	0.0003483	unsafe
6	30	140	200	200	500			0.0001844	0.0003284	**safe**

CHAPTER 8

CONCLUSIONS

1. The thickness of pavement varies with the change in the value of C.B.R. With higher value of C.B.R. the pavement thickness is less and vice versa. In this study we used 90th percentile existing soil CBR value 4.6% & Borrow soil CBR value 8.5%.

2. The average daily traffic is obtained from the traffic survey data: average traffic volume National Highway NH365BB (Suryapeta – Khammam) with chainage length of 10km from Mothey, Suryapet to Nayakulagudem, Kothagudem Telangana is 2460 CVPD.

3. The Pavment layer thickness for construction of 6-lane flexible pavement as per IRC 37 -2018 designed those thickness are ; SURFACE COURSE = 30mm, BASE / BINDER COURSE = 140mm, WMM = 200mm, GSB = 200mm, SUBGRADE = 500mm. Due to the saving in Pavement thickness is less quantity of material will be applicable so that, huge amount of money can be saved.

FUTURE SCOPE

1. The present study is made on one road and the pattern of deterioration is determined on basis of the analysis. However, the more realistic pattern of deterioration can be developed by studying the different road.
2. The study can also be with different bitumens grades, modified bitumen, RAP utilization.
3. The study can also be elaborated by varying the various factors such as temperature and Rainfall Data.
4. Further this research work can be carried with different materials to improve CBR values by soil stabilization technique.

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